

MA 220: Partial Differential Equations

Course Code: MA 220

Course Name: Partial Differential Equations

L-T-P-C: 3-1-0-4

Prerequisites: Computer Organization and Processor Architecture Design (EE 326) or equivalent.

Students intended for: B.Tech Maths and Computing

Elective or Core: Core for B. Tech Maths and Computing and Elective for other discipline

Mutual Exclusion: MA 522

Approval: 59th BoA

Course Contents

- **Unit 1:** Recall the definition of directional derivatives, chain rules and total derivatives, Introduction to PDE, First order PDEs, Solution methods for first order PDE. [5 Lectures]
- **Unit 2:** Classification of first order PDE, Solutions of linear and non-linear PDEs, Cauchy Problem, Cauchy Kowalevski Theorem. [5 Lectures]
- **Unit 3:** Classification of Second Order Partial Differential Equations: normal forms and characteristics. Initial and Boundary Value Problems: Green's identity. [7 Lectures]
- **Unit 4:** Methods of Solution, Methods of separation of variables, Characteristic method, Green's function, Fourier transform. [7 Lectures]
- **Unit 5:** Laplace equation: mean value property, weak and strong maximum principle, Green's function, Poisson's formula, Dirichlet's principle. [5 Lectures]
- **Unit 6:** Heat equation: initial value problem, fundamental solution, weak and strong maximum principle and uniqueness results. Application of Heat equation. [5 Lectures]
- **Unit 7:** Wave equation: uniqueness, D'Alembert's method, method of spherical means and Duhamel's principle. Application of Wave equation. [5 Lectures]

Text books:

1. E. C. Zachmanoglou and D. W. Thoe, **Introduction to Partial Differential Equations**, Dover Publications, New York, 2016.
2. L.C. Evans, **Partial Differential Equations**, Graduate Studies in Mathematics, Vol. 19, AMS, Providence, 1998

References:

1. G. B. Folland, **Introduction to Partial Differential Equations**, Princeton University Press, 1995
2. A. K. Nandakumaran and P. S. Datti, **Partial Differential Equations**, Cambridge-IISC Series, 2020.
3. W. E. Schiesser and G. W. Griffiths, **A Compendium of Partial Differential Equation Models**, Cambridge University Press, 2009.